less than a second ATB, then even though they are much closer, there is still no way for them to have any influence on each other since light would require a whole second to travel the distance between them.2 If in order for their separation to be much less, say 186 miles, we have to run the film back to less than a thousandth of a second ATB, then, again, the same conclusion follows: They can't influence each other since in less than a thousandth of a second light can't travel the 186 miles separating them. Carrying on in the same vein, if we have to run the film back to less than a billionth of a second ATB in order for these regions to be within one foot of each other, they still cannot influence each other since there is just not enough time since the bang for light to have traveled the 12 inches between them. This shows that just because two points in the universe get closer and closer as we head back to the bang, it is not necessarily the case that they can have had the thermal contact—like that between soup and air—necessary to bring them to the same temperature.

Physicists have shown that precisely this problem arises in the standard big bang model. Detailed calculations show that there is no way for regions of space that are currently widely separated to have had the exchange of heat energy that would explain their having the same temperature. As the word horizon refers to how far we can see—how far light can travel, so to speak—physicists call the unexplained uniformity of temperature throughout the vast expanse of the cosmos the "horizon problem." The puzzle does not mean the standard cosmological theory is wrong. But the uniformity of temperature does strongly suggest that we are missing an important part of the cosmological story. In 1979, the physicist Alan Guth, now of the Massachusetts Institute of Technology, wrote the missing chapter.

Inflation

The root of the horizon problem is that in order to get two widely separated regions of the universe close together, we have to run the cosmic film way back toward the beginning of time. So far back, in fact, that there is not enough time for any physical influence to have traveled from one region to the other. The difficulty, therefore, is that as we run the cosmological film backward and approach the big bang, the universe does not shrink at a fast enough rate.